Track 20: Embedded Systems

June 2026

Track 20: Embedded Systems

MR ROKESH TECHNOLOGY — Training & Internship Programs

> ***Skill path:*** *C/C++ → Microcontrollers → RTOS → Peripherals → Communication Protocols → Firmware Design*

Related: Track 22 — Arduino | Track 21 — Raspberry Pi | Track 19 — Robotics & IoT

Track Overview

Who this is for: ECE/EEE students and engineers targeting embedded firmware and hardware-software integration roles.

Prerequisites: C programming basics; digital electronics helpful.

Tools: ARM Cortex boards (STM32), Arduino, Keil/STM32CubeIDE, oscilloscope/logic analyzer (lab), UART/SPI/I2C, FreeRTOS.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Embedded awareness	GPIO + UART firmware project
1 Month	80–100	Junior embedded basics	Multi-peripheral firmware
3 Months	250–300	Internship-ready	RTOS-based device firmware
6 Months	500–600	Embedded engineer entry	Production firmware capstone
9 Months	750–900	Industry-ready	Certified-style portfolio + 2 capstones

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Embedded C, memory, bit manipulation	5 embedded C programs
D3–4	GPIO, timers, interrupts	LED blink + button interrupt
D5–6	UART, SPI, I2C communication	Sensor read via I2C
D7–8	ADC, PWM, motor control basics	DC motor speed control
D9–10	Integration	**Multi-peripheral firmware project**

1-Month Foundation

Week	Topics	Project
W1	C for embedded, registers, GPIO	Digital I/O controller
W2	Timers, interrupts, debouncing	Event-driven input system
W3	Serial protocols, sensor drivers	Environmental monitor
W4	State machines, low-power modes	**Battery-powered sensor node**

3-Month Job Ready

Month	Topics	Projects
M1	STM32/ARM setup, HAL, debugging	Blink + UART + ADC labs
M2	FreeRTOS, tasks, queues, semaphores	Multi-task firmware
M3	Bootloader intro, OTA concepts, capstone	**RTOS device firmware capstone**

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	CAN bus, Modbus, industrial protocols	Protocol bridge firmware
M5	Unit testing (Unity/CMock), code review	Tested driver library
M6	Capstone	**Production embedded product firmware**

9-Month Advanced

Capstones: IoT gateway firmware + safety-critical control module | **Level:** L5

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	C, peripherals, RTOS, protocols
Lab Completion	40%	Firmware compiles; hardware behaves correctly
Project	30%	Code quality, documentation, schematics
Portfolio	10%	GitHub, demo videos, design docs

Appendix: Mentor Notes

Hardware: STM32 Nucleo or equivalent recommended for 3-month+ tiers. Always verify pin configs before powering circuits.